

PROFESSIONAL SUMMARY

Mechatronics Engineering graduate with hands-on experience in embedded systems, IoT, automation, PCB design, and field validation. Skilled in ESP32, Raspberry Pi, sensor integration, actuator control, Python, KiCad, and prototype testing, with practical internship experience in automated irrigation and water monitoring systems.

TECHNICAL SKILLS

Programming:	Python, MATLAB, Arduino/C++
Microcontrollers:	ESP32, Arduino, Raspberry Pi
Embedded & IoT:	Sensor Interfacing, MOSFET Switching, Flow Sensors, LoRaWAN, TTN, Wi-Fi
Circuit & PCB Design:	KiCad, Power Regulation, Relay/MOSFET Circuits, Solenoid Valve Control
Software & Backend:	Python, MySQL, HTML, CSS, JavaScript, REST API Basics
Mechanical Design:	Fusion 360, 3D Printing, Enclosure Design, Prototype Testing
Tools:	Proteus, Simulink, ABB RobotStudio, MS Office, ROS basics

EDUCATION

- B.E. in Mechatronics Engineering (Honours) | Visvesvaraya Technological University 2022 – 2026
CGPA: 8.8
- Cambridge International AS & A Levels | Budhanilkantha School 2020 – 2022
Subjects: Physics, Chemistry, Computer Science, Mathematics, English General Paper

WORK EXPERIENCE

Embedded and Control Systems Intern | Gridomatix – Bangalore, India Jan 2026 – Present

- Engineered and tested ESP32-based embedded circuits for an automated irrigation system using solenoid valves, MOSFET switching, flow sensors, relays, and power regulation modules.
- Implemented embedded control logic using Arduino/C++ for actuator control, sensor reading, system response, and hardware validation.
- Created circuit schematics and PCB layouts in KiCad, and developed Python-MySQL backend logging workflows for device data, sensor logs, and irrigation summaries. 3+ PCB design and prototype iterations.

Python Web Development Intern | e-Zone International Pvt. Ltd. – Kathmandu, Nepal Feb 2024 – Apr 2024

- Built responsive landing pages for digital marketing campaigns and collaborated with UI/UX and marketing teams to improve user engagement and campaign performance by 37%.

Teaching Instructor | Next Generation Residential Academy, Kathmandu, Nepal July 2022 – Sept 2022

- Taught science, mathematics, and computer science to middle and high school students through interactive learning methods.

PROJECTS

Automated Irrigation Control System, Water Monitoring System / Gridomatix

- Developed an ESP32-based irrigation and water monitoring system with flow sensing, solenoid control, PCB design, data logging, enclosure development, and pilot-site validation. Reduced manual intervention upto 90% and validated operation across 20+ hours of bench and field testing.

Autonomous Lawn Mower using Raspberry Pi, Camera and GPS

- Designed an autonomous robotic lawn mower using Raspberry Pi, camera-based grass detection, GPS, proximity sensors, and Python/OpenCV-based zig-zag path planning. 94% average lawn coverage and 95% obstacle avoidance. Completed mowing cycles in 6-7 minutes on 14-16 m² test area.

IoT-Based Automatic Water Level Controller using ESP32

- Integrated an ESP32-based water-level monitoring and pump-control system with sensors, relay switching, OLED display, Wi-Fi monitoring, and automatic pump safety logic.

Miniature Home Automation System using Arduino

- Built an Arduino-based home automation prototype using PIR, temperature, and LDR sensors with relay-based appliance control and event-based safety responses.

PUBLICATIONS AND PATENTS

- S. Bhattarai et al., "IoT Driven Home Automation System: A Comprehensive Review," IJERT, Vol. 13, No. 3, Apr. 2025. DOI: 10.17577/IJERTCONV13ISO3017
- Presented "Vision-Guided Robotic Lawn Mower for Autonomous Lawn Maintenance" at ISC-SAI 2025; manuscript currently under publication process.
- Design Patent Applied: Application No. 484839-001, CBR No. 228436